COURIER MANAGEMENT SYSTEM

2. REQUIREMENT ANALYSIS

2.1 Project Overview

Requirement analysis defines what the Courier Management System must provide and the quality characteristics it should maintain. The system is designed to manage customer, courier, shipment and delivery information through structured data management, validation, tracking, reports and dashboards.

2.2 Functional Requirements

ID	Requirement	Description
FR-1	User Authentication	Provide secure access to authorized users
FR-2	Farm, Crop, Farmer and Buyer Management	Create, maintain and manage customer and courier records..
FR-3	Data Entry and Validation	Capture required shipment information and validate entered values.
FR-4	Relationship Mapping	Track courier shipments using unique tracking information and status updates.
FR-5	Report Generation	Maintain pickup, delivery and shipment status information..
FR-6	Dashboard Visualization	Generate reports for courier bookings, shipments, deliveries and customer information.
FR-7	User Role Management	Display important courier and shipment information using dashboards and visual analytics..

2.3 Non-Functional Requirements

Category	Requirement
Usability	Provide a simple and user-friendly interface for managing courier operations.
Security	Use authentication and access controls for protecting courier and customer information.
Reliability	Maintain accurate shipment records and reduce the risk of data loss.
Performance	Target report, search and shipment information loading within approximately 3 seconds.

Availability	Provide access to the system for authorized users whenever required
Scalability	Support growth in customers, courier shipments, delivery records and system users.

2.4 Data Flow Diagram (Logical Flow)

User → Courier Management System UI → Customer/Courier/Shipment/Delivery Records → Validation → Shipment Tracking & Status Updates → Reports → Dashboards

2.5 Technology Stack

Layer	**Technology**
Frontend	User interface/Salesforce Lightning UI
Backend/Application Platform	Salesforce Platform
Database	Salesforce Cloud
Automation/Validation	Salesforce Validation Rules and automation capabilities
Analytics	Salesforce Reports and Dashboards

2.6 Main Data Entities

- Customer: Stores customer identity, contact and address information.
- Courier: Stores courier booking, parcel and shipment information.
- Shipment: Stores tracking number, pickup details, delivery details and shipment status.
- Delivery: Stores delivery information, delivery status and completion details.

2.7 Requirement Summary

The requirements establish a centralized, secure and structured Courier Management System with customer and shipment management, data validation, tracking, delivery updates and analytics. These requirements guide the design and implementation phases.